HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Tracked emerging AI models and techniques, translating research into actionable insights and presenting a curated shortlist of relevant technologies for business application.
- Developed and executed small-scale prototypes against internal business challenges, delivering clear 'keep-or-kill' recommendations based on empirical testing results.
- Managed the transition of successful AI pilot concepts into production-ready solutions, collaborating with the Digital team and operational stakeholders for seamless deployment and handover.
- Designed and implemented Python-based automation scripts to streamline data ingestion and ETL processes, reducing manual data preparation effort by an estimated **30%** for reporting teams.
- Built and maintained **Java Spring Boot** REST APIs for inter-service communication, ensuring adherence to established coding standards and API design principles.
- Containerized multiple Java applications using **Docker** and deployed them to **AWS EC2** instances, improving deployment consistency and simplifying environment management.
- Authored detailed technical documentation and runbooks for newly deployed AI solutions, facilitating user adoption and enabling day-to-day operational support by relevant teams.

SKILLS

AI/ML Concepts – LLM/SLM landscape, generative AI, model prototyping, proof of concept, AI research, AI trends

Programming Languages – Python, Java, SQL, R

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, LangChain, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Kubernetes

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, data modelling, data warehousing

Databases – PostgreSQL, MySQL, Oracle

PROJECTS

AI-Driven Document Summarization Prototype (HCL Technologies)

- Developed a prototype using **Python** and **LangChain** to ingest and summarize lengthy internal business reports, aiming to reduce reading time for leadership.
- Evaluated several pre-trained language models (e.g., open-source alternatives) to identify the most effective approach for generating concise and accurate summaries, reaching a qualitative assessment of **85% coherence**.
- Packaged the summarization logic into a simple **FastAPI** service, demonstrating its potential integration into existing document management workflows.

Predictive Maintenance Analytics Dashboard (MSc Project)

- Engineered data pipelines using **Python** and **pandas** to extract and clean time-series sensor data from a simulated manufacturing environment.
- Built interactive **Power BI** dashboards visualizing key performance indicators and predicted equipment failure probabilities, enabling proactive maintenance scheduling for an **8-person** operations team.
- Applied fundamental **machine learning** concepts to build a basic predictive model, achieving an initial **~0.72 AUC** on identifying patterns preceding simulated failures.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)